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MLDL Experiment 11

Lung Cancer Prediction using Logistic Regression

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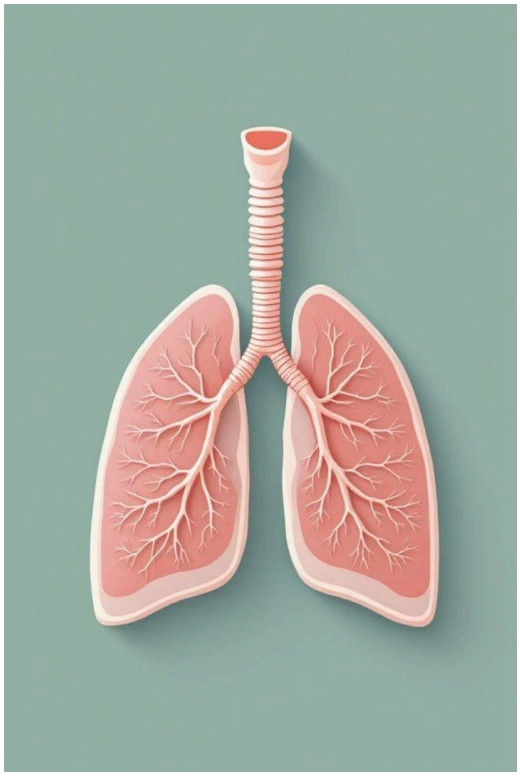


Introduction & Problem Statement

What is Lung Cancer?

Lung cancer is one of the most prevalent and deadly cancers globally, characterised by uncontrolled cell growth in lung tissue. Early detection significantly improves treatment outcomes and survival rates.

Problem Statement

Develop a classification model to predict lung cancer presence using patient health data. This addresses the critical need for faster, cost-effective diagnosis tools in healthcare settings.

Objectives & Motivation

Primary Objective

Build a classification model using Logistic Regression to predict binary outcome (0 = No Cancer, 1 = Cancer)

Motivation

Faster, cost-effective diagnosis enabling early intervention and improved patient outcomes

Expected Outcomes

Accurate prediction model with interpretable results for clinical decision support

Literature Review & Proposed System

01

Existing Models

Decision Trees, Support Vector Machines, Neural Networks have been applied to cancer prediction with varying success

02

Why Logistic Regression?

Simple, interpretable, computationally efficient, provides probability estimates, and handles linear relationships well

03

Proposed System Flow

Input Data → Preprocessing → Model Training → Prediction Output

Dataset & Features

Dataset Details

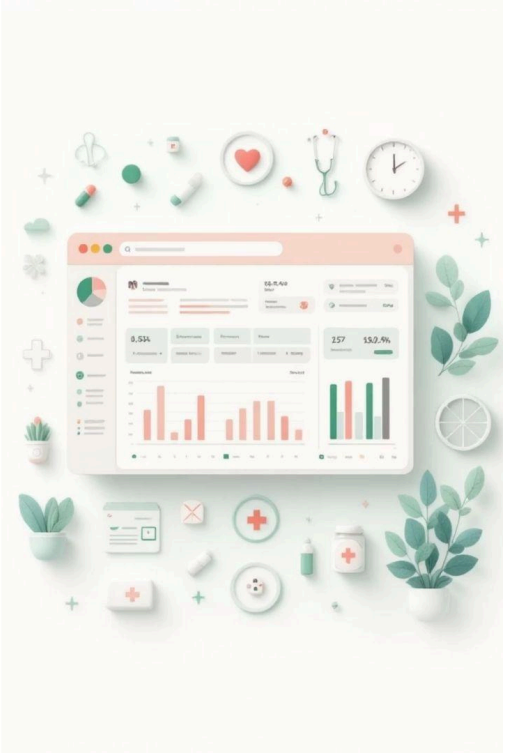
The Lung Cancer Dataset contains patient health records with multiple risk factors. Target variable 'Level' categorises risk as Low, Medium, or High.

Feature Engineering

Conversion: Low risk = 0, Medium/High risk = 1 (binary classification)

Key Features

- Age, Gender
- Smoking habits
- Anxiety levels
- Peer pressure
- Chronic disease
- Fatigue, Snoring



Data Preprocessing & Visualization

Data Cleaning
Removed unnecessary columns, checked for missing values, ensured data quality

Label Encoding
Converted categorical variables to numerical format for model compatibility

Visual Analysis
Generated histograms for feature distributions and correlation heatmap for relationships

Methodology & Model Details

1

Data Preparation
Dataset loading and feature selection

2

Train-Test Split
80:20 ratio for validation

3

Model Training
Logistic Regression with lbfgs solver

4

Prediction
Binary classification output

Algorithm Parameters

- **Algorithm:** Logistic Regression
- **Solver:** lbfgs
- **Max iterations:** 2000

Implementation Details

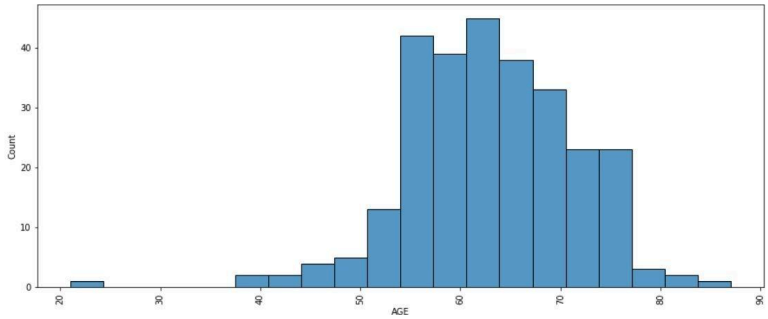
Model trained using scikit-learn library with appropriate hyperparameters for convergence.

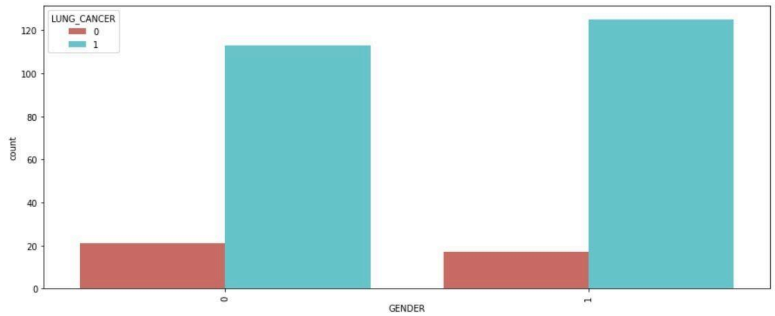
```
data.head()
```

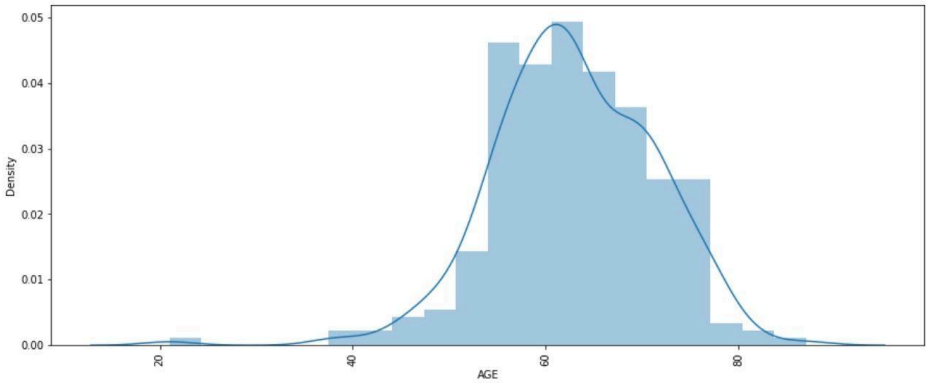
	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL CONSUMING	COUGHING	SHORTNESS OF BREATH	SWALLOWING DIFFICULTY	CHEST PAIN	LUNG_CANCER
0	M	69	1	2	2	1	1	2	1	2	2	2	2	2	2	YES
1	M	74	2	1	1	1	2	2	2	1	1	1	2	2	2	YES
2	F	59	1	1	1	2	1	2	1	2	1	2	2	1	2	NO
3	M	63	2	2	2	1	1	1	1	1	2	1	1	2	2	NO
4	F	63	1	2	1	1	1	1	1	2	1	2	2	1	1	NO

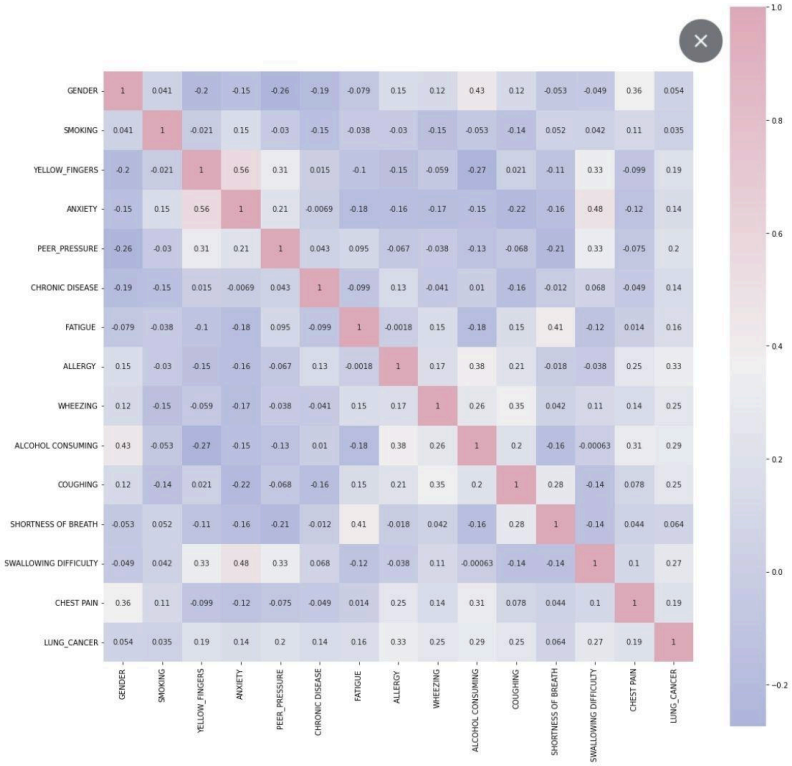
```
data.tail()
```

	GENDER	AGE	SMOKING	YELLOW_FINGERS	ANXIETY	PEER_PRESSURE	CHRONIC DISEASE	FATIGUE	ALLERGY	WHEEZING	ALCOHOL CONSUMING	COUGHING	SHORTNESS OF BREATH	SWALLOWING DIFFICULTY	CHEST PAIN	LUNG_CANCER
304	F	56	1	1	1	2	2	2	1	1	2	2	2	2	1	YES
305	M	70	2	1	1	1	1	2	2	2	2	2	2	1	2	YES
306	M	58	2	1	1	1	1	1	2	2	2	2	1	1	2	YES
307	M	67	2	1	2	1	1	2	2	1	2	2	2	1	2	YES
308	M	62	1	1	1	2	1	2	2	2	2	1	1	2	1	YES






0.05
0.04
0.03
0.02
0.01
0.00
Density
20
40
60
80
AGE


GENDER 1 0.041 -0.2 -0.15 -0.26 -0.19 -0.079 0.15 0.12 0.43 0.12 -0.053 -0.049 0.36 0.054
SMOKING 0.041 1 -0.021 0.15 -0.03 -0.15 -0.038 -0.03 -0.15 -0.053 -0.14 0.052 0.042 0.11 0.035
YELLOW_FINGERS -0.2 -0.021 1 0.56 0.31 0.015 -0.1 -0.15 -0.059 -0.27 0.021 -0.11 0.33 -0.099 0.19
ANXIETY -0.15 0.15 0.56 1 0.21 -0.0069 -0.18 -0.16 -0.17 -0.15 -0.22 -0.16 0.48 -0.12 0.14
PEER_PRESSURE -0.26 -0.03 0.31 0.21 1 0.043 0.095 -0.067 -0.038 -0.13 -0.068 -0.21 0.33 -0.075 0.2
CHRONIC DISEASE -0.19 -0.15 0.015 -0.0069 0.043 1 -0.099 0.13 -0.041 0.01 -0.16 -0.012 0.068 -0.049 0.14
FATIGUE -0.079 -0.038 -0.1 -0.18 0.095 -0.099 1 -0.0018 0.15 -0.18 0.15 0.41 -0.12 0.014 0.16
ALLERGY 0.15 -0.03 -0.15 -0.16 -0.067 0.13 -0.0018 1 0.17 0.38 0.21 -0.018 -0.038 0.25 0.33
WHEEZING 0.12 -0.15 -0.059 -0.17 -0.038 -0.041 0.15 0.17 1 0.26 0.35 0.042 0.11 0.14 0.25
ALCOHOL CONSUMING 0.43 -0.053 -0.27 -0.15 -0.13 0.01 -0.18 0.38 0.26 1 0.2 -0.16 -0.00063 0.31 0.29
COUGHING 0.12 -0.14 0.021 -0.22 -0.068 -0.16 0.15 0.21 0.35 0.2 1 0.28 -0.14 0.078 0.25
SHORTNESS OF BREATH -0.053 0.052 -0.11 -0.16 -0.21 -0.012 0.41 -0.018 0.042 -0.16 0.28 1 -0.14 0.044 0.064
SWALLOWING DIFFICULTY -0.049 0.042 0.33 0.48 0.33 0.068 -0.12 -0.038 0.11 -0.00063 -0.14 -0.14 1 0.1 0.27
CHEST PAIN 0.36 0.11 -0.099 -0.12 -0.075 -0.049 0.014 0.25 0.14 0.31 0.078 0.044 0.1 1 0.19
LUNG_CANCER 0.054 0.035 0.19 0.14 0.2 0.14 0.16 0.33 0.25 0.29 0.25 0.064 0.27 0.19 1
GENDER
SMOKING
YELLOW_FINGERS
ANXIETY
PEER_PRESSURE
CHRONIC DISEASE
FATIGUE
ALLERGY
WHEEZING
ALCOHOL CONSUMING
COUGHING
SHORTNESS OF BREATH
SWALLOWING DIFFICULTY
CHEST PAIN
LUNG_CANCER
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

Results & Performance Graphs

89%
Accuracy
Correct predictions

0.87
Precision
Positive predictions

0.83
Recall
Sensitivity

0.88
Specificity
True negative rate

Performance, Advantages & Challenges

Advantages

- Simple and fast implementation
- Highly interpretable results
- Provides probability estimates
- Low computational cost

Challenges

- Feature mismatch (missing Snoring in test data)
- Encoding consistency issues
- Single-class test data limitation
- Assumes linear relationships

Applications, Future Scope & Conclusion

Current Applications

Healthcare diagnosis support, early cancer detection screening, clinical decision assistance

Future Enhancements

Implement Deep Learning architectures, expand dataset diversity, incorporate imaging data integration

Conclusion

Logistic Regression demonstrates effectiveness for early lung cancer prediction with interpretable results suitable for clinical settings

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Thank You

Questions & Discussion